



**Politecnico
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Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

Group 1

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CHAPTER 1

Intro

The device investigated in this laboratory is a double quantum dot based on a Si-SiO₂ heterostructure. Because of the difference between material band gaps, a discontinuity within the conduction and valence bands is created. By engineering the structure in a specific manner, it becomes feasible to confine carriers (electrons and holes) along the growth direction.

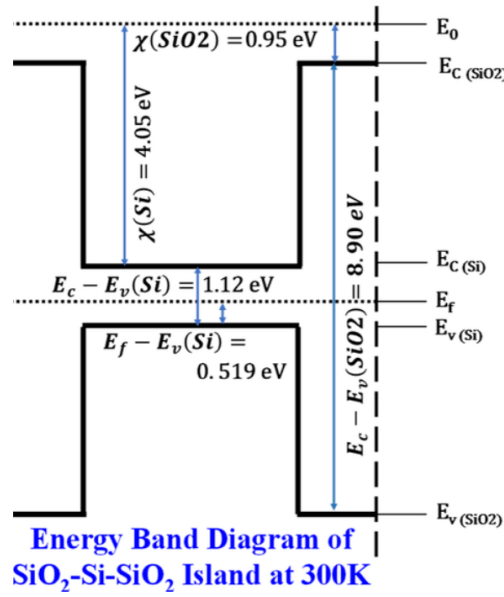


Figure 1.1: Type I heterostructure Si-SiO₂

The figure shows the band diagram of this type of heterostructure. The quantum well obtained in this way allows for confinement along the growth direction, thus creating a 2D Electron Gas. Further confinement along the other two directions is achieved thanks to the application of particular potentials by various types of gates.

- The plunger gates control the electron occupation number of each dot by shifting the energy levels up or down;
- the barrier gates control the tunnel couplings between the two dots and between each dot and its respective lead (source or drain);

- the Y-gate is a secondary tuning electrode that complements the main control from the plunger and barrier gates by refining the potential profile of the entire dot region.

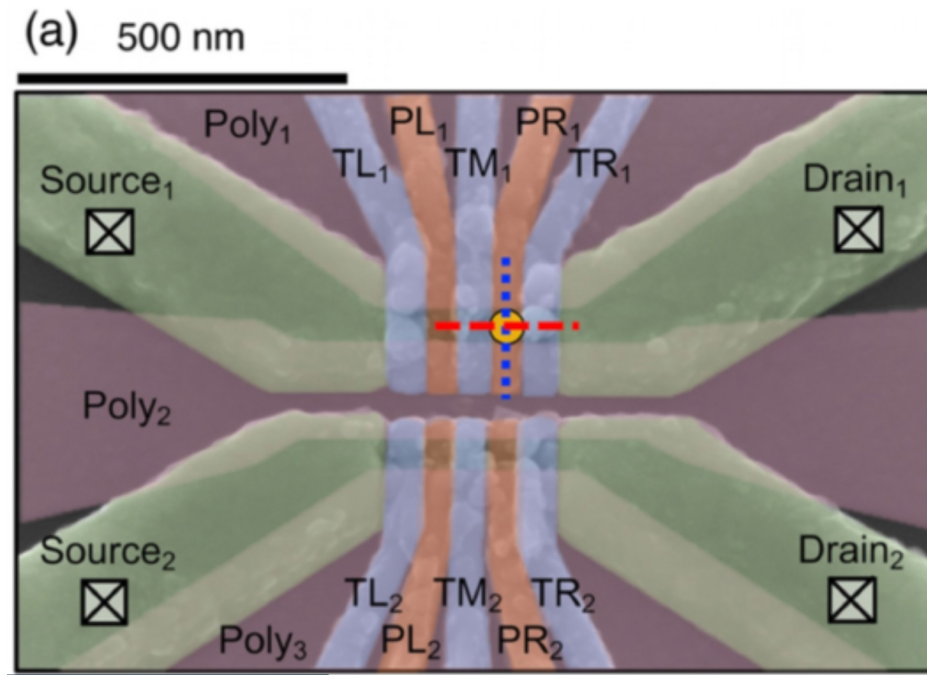


Figure 1.2

CHAPTER 2

Mesh

The dimensions of the device are 353 nm along the x direction, 273 nm along y and 118 nm along z .

Question: Determine the regions exhibiting greater density. Analyze various sets of regions grouped by "`region_name_region`". Focus on regions labeled with names starting with "`region _ quantum`". Identify specific indicators of this varying density within the code, corresponding to the graph recently examined.

The distance between the points of the mesh is controlled by `grid_spacing_0X`, a parameter where the smaller the number X, the tighter the mesh and thus the higher the density. The smallest parameters appear in the definition of:

- `region_pre_quantum_A/B/C;`
- `region_quantum_layer;`
- `region_Y_gate;`
- `region_Reservoir_gate;`
- `region_Barrier_gate;`
- `region_Plunger_gate.`

We can conclude that the mesh is tighter for the quantum regions, the doped interfaces and the gates.

CHAPTER 3

Poisson

Question: Determine the applied voltages across distinct regions and comprehend the operational temperature.

voltages are defined in this piece of code:

```
V_Y_gate_1 = 0.535
V_Y_gate_2 = 0.535
V_region_Reservoir_Left = 1.3      # Left reservoir voltage [V]
V_region_Reservoir_Right = 1.3     # Right reservoir voltage [V]
V_Barrier_1 = 0.040                # Left barrier voltage [V]
V_Barrier_2 = 0.4                  # Interdot barrier voltage [V]
V_Barrier_3 = 0.04                 # Right barrier voltage [V]
V_Plunger_Left = 0.70              # Left dot voltage [V]
V_Plunger_Right = 0.70+0.0001990577078 # Right dot voltage
```

The operational temperature is set to 15 mK.

Question: Explain the different materials listed in the code.

The materials used are Silicon for the bulk and quantum regions, SiO₂ for the oxide layers and Al₂O₃ for the gates.

Question: Report what the parameters set in the following code represent.

```
params_poisson = PoissonSolverParams()
params_poisson.tol = 1e-7
params_poisson.initial_ref_factor = 0.55
params_poisson.final_ref_factor = 0.85
params_poisson.min_nodes = 0
params_poisson.maxiter = 1000
params_poisson.dot_region = dot_region
params_poisson.h_dot = 0.3
params_poisson.max_nodes = 1e10
```

The parameters set in the provided Python code configure a Poisson solver.

- `tol = 1e-7`: tolerance in Volts for convergence. The iterative solver will stop once the residual falls below this threshold. Default is `1e-6`.

Note: with that value of the tolerance set, the code required a lot of memory and the server was overloaded. We were therefore forced to reduce the value of tolerance to `1e-3`. The output was courtesy of the group 3. The simulation produced the following messages:

```
Error converged below required tolerance after:  
60 Poisson iterations. > 12330.196757555008 s.
```

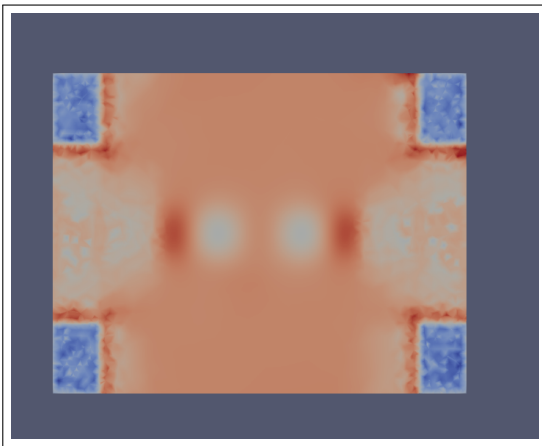
```
Average computation time / Poisson iteration is 205.5032798131307 s.
```

- `initial_ref_factor = 0.55`: it is a float between 0 and 1. It represents the maximal relative error threshold used at every adaptive iteration. The default is 0.1.
- `final_ref_factor = 0.85`: it is a float between 0 and 1. It represents the factor that relates to the 'amount of mesh refinement' that will occur once the solver saturates. The lower the number the more refined the output mesh will be. Default is set at 0.75.
- `min_nodes = 0`: Minimal number of nodes allowed by the solver. If Poisson solver converges with less nodes, then the initial mesh will be uniformly refined and the adaptive procedure will restart. Default is `1e5`.
- `maxiter = 1000`: Maximum number of iterations in a self-consistent loop. Default is 200.
- `dot_region`: list of any combination of strings and lists of two 3-tuples or a single list of two 3-tuples or a single string. Regions where we would like to set the maximal value of the characteristic length. These regions are identified either by their physical name or by a list of two 3-tuples, specifying the minimum and maximum (x, y, z) coordinates of the two opposite corners of a box containing the region. Defaults to None, in which case no maximal characteristic length will be set in any region.
- `h_dot`: it is a float or list of floats. Maximal value of the characteristic length in each of the regions defined in refined region. If a list, the index of each element must correspond to the index of the region in refined region. If a single float, the same value is used for all regions.
- `max_nodes = 1e10`: maximal number of nodes permitted for an adaptive iteration to be performed. If the maximum is exceeded no additional adaptive iterations will take place. Default is set at `1e6`.

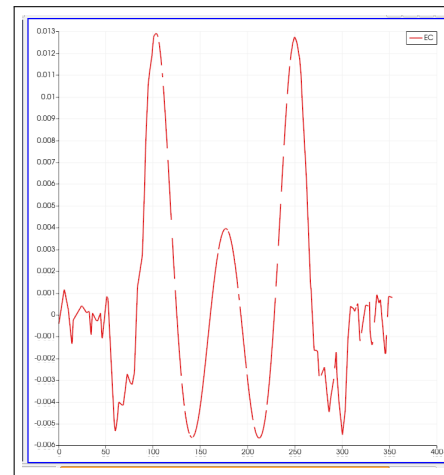
CHAPTER 4

Paraview

Question: Explain the areas present in the cut and the associated colors. Click on "Plot Overline," and vary the variable x across the domain, with y values set to 136.5 and z remaining at 45. Describe the graph, showing how the potential changes in the domain and what the minima represent. Explain if this behavior in the central region is as expected.



(a)



(b)

Image (a) shows a color map representing the potential across this slice of the device. White indicates zero potential, with increasingly blue shades corresponding to more negative values and increasingly red shades to more positive ones.

The four blue regions at the corners of the device are the source and drain contacts. As expected, since they serve as reservoirs of electrons, they exhibit a negative potential.

At the extremities of the device, in light blue, we can find the characteristic Y-gates, that connect the tunnel gates to source and drain.

The dark red dots correspond to the external tunnel gates, at higher potential to isolate the dots from the source and drain regions. The lighter red at the center is the tunnel gate that controls the passage of the electrons between the two dots, represented by the two other light blue regions.

The line plot confirms our observations from the slice. The two peaks correspond to the potentials of the tunnel gates, while the two minima indicate the regions influenced by the plunger gates, where

the electrons are confined within the quantum dots.